

COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	08/27/2026	DATE SUBMITTED:	08/29/2026
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ACTIVITY NO. 5.1 Consolidating Playbook plays

1. PROCEDURE OUTPUT

Server 1 Ubuntu1	192.168.56.118
Server 2 Ubuntu2	192.168.56.119
Server 3 CentOS	192.168.56.117

TASK 1:

- In the local machine, make sure you are in the local repository directory (CPE232_yourname). Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issued this command. Did something happen? Why? **It pulled my previous commits such as HOA4. It didnt prompt a passphrase and i was able to connect to my repository with the ssh key.**

```

fegalanvm@fegalan-workstation:~/CPE212_fegalan$ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 4 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 459 bytes | 459.00 KiB/s, done.
From github.com:christopherfegalan/CPE212_fegalan
  805d840..045cbec  main    -> origin/main
Updating 805d840..045cbec
Fast-forward
 HOA4/install_apache.yaml | 17 ++++++
 1 file changed, 17 insertions(+)
 create mode 100644 HOA4/install_apache.yaml
fegalanvm@fegalan-workstation:~/CPE212_fegalan$ ls
HOA4  README.md
fegalanvm@fegalan-workstation:~/CPE212_fegalan$

```

```
fegalanvm@fegalan-workstation:~/CPE212_fegalan$ ansible-playbook --ask-become-pass install_apache.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
fatal: [192.168.56.117]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: fegalanvm@192.168.56.117: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).", "unreachable": true}
ok: [192.168.56.119]
ok: [192.168.56.118]

TASK [update repository index] *****
```

```
fegalanvm@fegalan-workstation: ~/CPE212_fegalan
GNU nano 6.2 install_apache.yaml
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
    when: ansible_distribution == "Ubuntu"
```

[Wrote 19 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

```
fegalanvm@fegalan-workstation:~/CPE212_fegalan$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.117 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.117]
ok: [192.168.56.118]
ok: [192.168.56.119]

TASK [update repository index] *****
skipping: [192.168.56.117]
changed: [192.168.56.119]
changed: [192.168.56.118]

TASK [install apache2 package] *****
skipping: [192.168.56.117]
ok: [192.168.56.119]
ok: [192.168.56.118]

TASK [add PHP support for apache] *****
skipping: [192.168.56.117]
ok: [192.168.56.119]
ok: [192.168.56.118]

PLAY RECAP *****
192.168.56.117      : ok=1    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.118      : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
192.168.56.119      : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

fegalanvm@fegalan-workstation:~/CPE212_fegalan$
```

Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

- It skipped using the ip of centos because we specified to only run on ubuntu distribution.

```

fegalanvm@fegalan-workstation:~/CPE212_fegalan$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.119]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.117 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.117]
ok: [192.168.56.118]

TASK [update repository index] *****
skipping: [192.168.56.117]
changed: [192.168.56.119]
changed: [192.168.56.118]

TASK [install apache2 package] *****
skipping: [192.168.56.117]
ok: [192.168.56.119]
ok: [192.168.56.118]

TASK [add PHP support for apache] *****
skipping: [192.168.56.117]
ok: [192.168.56.118]
ok: [192.168.56.119]

TASK [update repository index] *****
skipping: [192.168.56.118]
skipping: [192.168.56.119]
ok: [192.168.56.117]

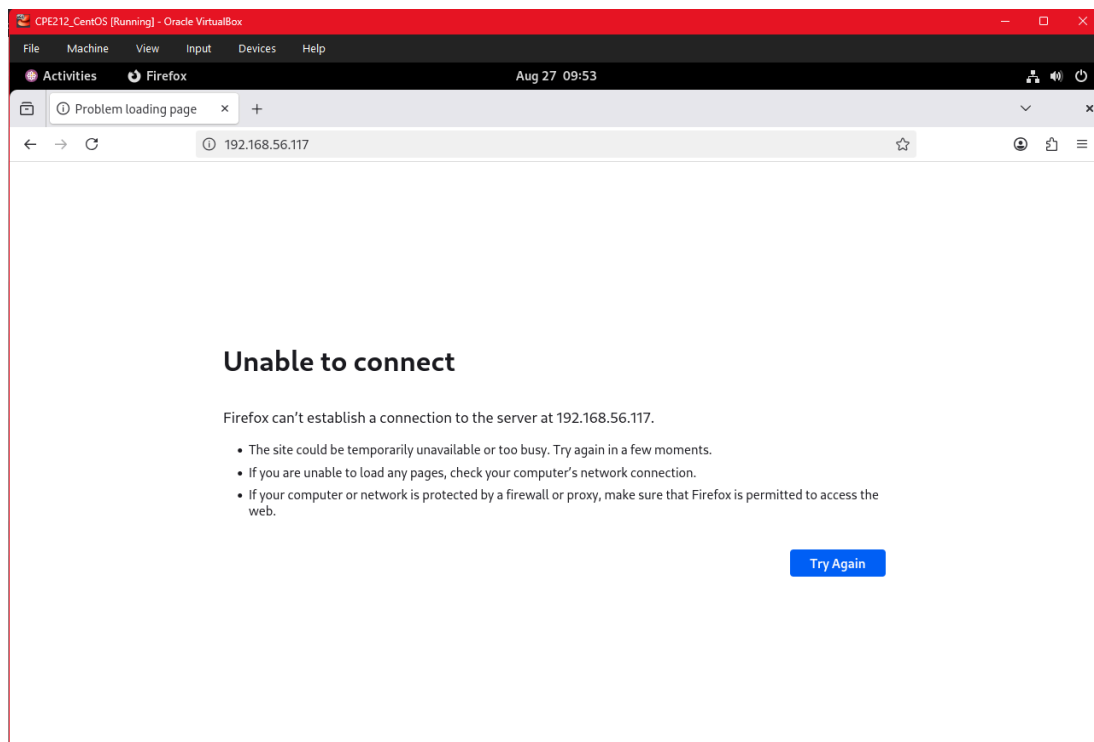
TASK [install apache2 package] *****
skipping: [192.168.56.118]
skipping: [192.168.56.119]
changed: [192.168.56.117]

TASK [add PHP support for apache] *****
skipping: [192.168.56.118]
skipping: [192.168.56.119]
changed: [192.168.56.117]

PLAY RECAP *****
192.168.56.117      : ok=4    changed=2    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.118      : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.119      : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
  
```

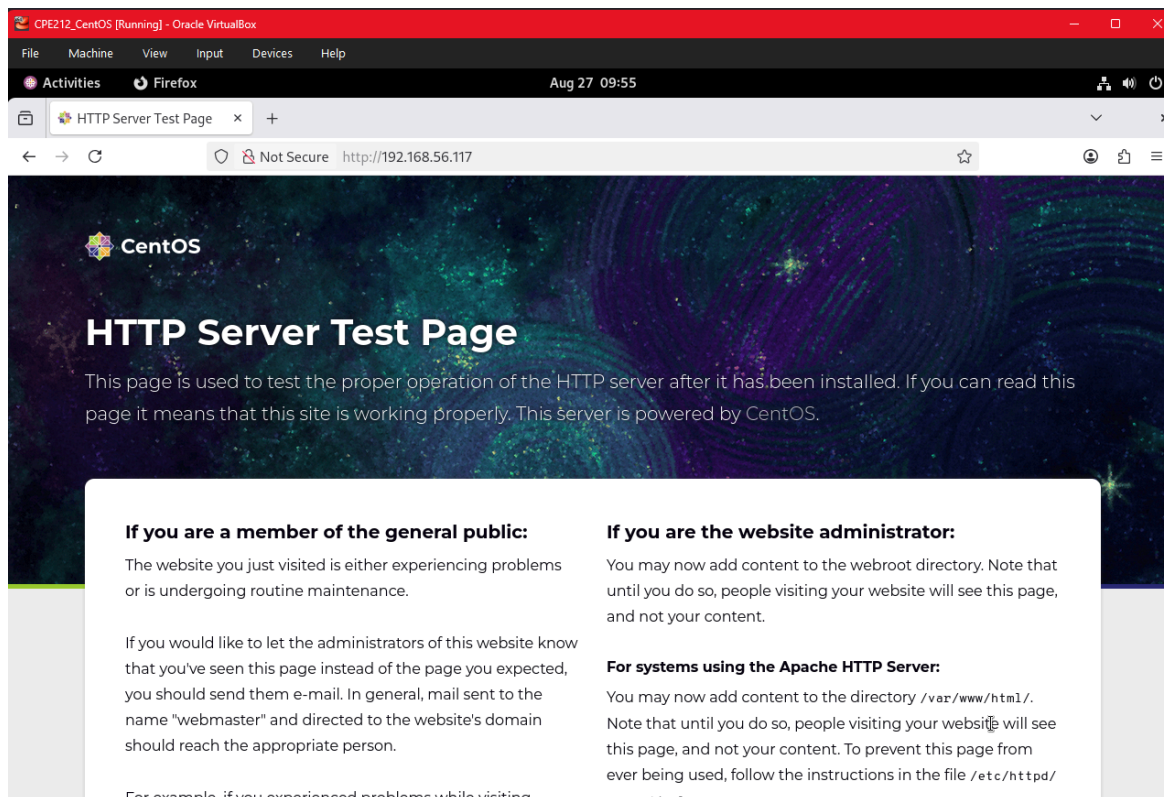
Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the result

- Created a way to run CentOS with the commands by running the same commands but instead using `dnf` for the CentOS distribution which skipped the ubuntu remote servers.



```
centos_chris@centoschris:~
[centos_chris@centoschris ~]$ systemctl status httpd
○ httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: d
   Drop-In: /usr/lib/systemd/system/httpd.service.d
           └─php-fpm.conf
   Active: inactive (dead)
   Docs: man:httpd.service(8)

[1]+  Stopped                  systemctl status httpd
[centos_chris@centoschris ~]$ sudo systemctl start httpd
[sudo] password for centos_chris:
[centos_chris@centoschris ~]$ sudo firewall-cmd --add-port=80/tcp
success
[centos_chris@centoschris ~]$
```



TASK 2:

```
--
- hosts: all
  become: true
  tasks:

    - name: update repository index for Ubuntu
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: update repository index for CentOS
      dnf:
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
```

[Read 30

```
TASK [update repository index for Ubuntu] *****
skipping: [CentOS]
changed: [Ubuntu1]
changed: [Ubuntu2]

TASK [install apache2 and php packages for Ubuntu] *****
skipping: [CentOS]
ok: [Ubuntu2]
ok: [Ubuntu1]

TASK [update repository index for CentOS] *****
skipping: [Ubuntu1]
skipping: [Ubuntu2]
ok: [CentOS]

TASK [install apache and php packages for CentOS] *****
skipping: [Ubuntu1]
skipping: [Ubuntu2]
ok: [CentOS]

PLAY RECAP *****
CentOS      : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
Ubuntu1    : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
Ubuntu2    : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

fegalanvm@fegalan-workstation:~/CPE212_fegalan$
```

Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the result.

- It basically did the same thing as the previous changes. But this time there were some changes made to the yaml file that condensed and removed the repetitive downloading of packages.

```

fegalanvm@fegalan-workstation:~/CPE212_fegalan$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [Ubuntu2]
ok: [Ubuntu1]
[DEPRECATION WARNING]: Distribution centos 9 on host CentOS should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using
the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This
feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in
ansible.cfg.
ok: [CentOS]

TASK [install apache2 and php packages for Ubuntu] *****
skipping: [CentOS]
ok: [Ubuntu1]
ok: [Ubuntu2]

TASK [install apache and php packages for CentOS] *****
skipping: [Ubuntu1]
skipping: [Ubuntu2]
ok: [CentOS]

PLAY RECAP *****
CentOS      : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
Ubuntu1     : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
Ubuntu2     : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

```

Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

- It removed the task of updating repository index and included it with the installing of packages using update_cache: yes.

```

TASK [Gathering Facts] *****
ok: [Ubuntu2]
ok: [Ubuntu1]
[DEPRECATION WARNING]: Distribution centos 9 on host CentOS should use /usr/libexec/platform-py
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible releas
the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprec
ansible.cfg.
ok: [CentOS]

TASK [install apache and php] *****
fatal: [Ubuntu1]: FAILED! => {"msg": "The task includes an option with an undefined variable. T
ckage' is undefined\n\nThe error appears to be in '/home/fegalanvm/CPE212_fegalan/install_apach
S, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending li
- name: install apache and php\n    ^ here\n"}
fatal: [Ubuntu2]: FAILED! => {"msg": "The task includes an option with an undefined variable. T
ckage' is undefined\n\nThe error appears to be in '/home/fegalanvm/CPE212_fegalan/install_apach
S, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending li
- name: install apache and php\n    ^ here\n"}
fatal: [CentOS]: FAILED! => {"msg": "The task includes an option with an undefined variable. Th
ckage' is undefined\n\nThe error appears to be in '/home/fegalanvm/CPE212_fegalan/install_apache
, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending lin
- name: install apache and php\n    ^ here\n"}

```

Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

- It was unsuccessful. The variables have no way of knowing which to use.

```

CPE212_Workstationn (SaveState Before CentOS SSH) [Running] - Oracle VirtualBox
Aug 27 10:14
fegalanvm@fegalan-workstation: ~/CPE212_fe
GNU nano 6.2 inventory.yaml
all:
  children:
    ubuntu:
      vars:
        ansible_user: fegalanvm
      hosts:
        server1:
          ansible_host: 192.168.56.118
          os_type: Ubuntu
          apache_package: apache2
          php_package: libapache2-mod-php
        server2:
          ansible_host: 192.168.56.119
          os_type: Ubuntu
          apache_package: apache2
          php_package: libapache2-mod-php
    centos:
      vars:
        ansible_user: centos_chris
      hosts:
        server3:
          ansible_host: 192.168.56.117
          os_type: CentOS
          apache_package: httpd
          php_package: php

fegalanvm@fegalan-workstation:~/CPE212_fegalan$ ansible-playbook -i inventory.yaml --ask-become-pass install_apache.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using
the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This
feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in
ansible.cfg.
ok: [server3]

TASK [install apache and php] *****
ok: [server1]
ok: [server3]
ok: [server2]

PLAY RECAP *****
server1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

- It resulted in ok state because it was already installed. Basically doing nothing and also by setting the packages on the inventory it was able to identify which package identifier should be used on each of the variables based on what OS is used.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

- Create a playbook that could do the previous tasks in Red Hat OS.

HOME:

Server 1 Ubuntu1	192.168.56.101
Server 2 Ubuntu2	192.168.56.104

Settings

Search

Type to search

christopherfegalan (christopherfegalan)

Your personal account

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Credentials

Organizations

Enterprises

SSH keys

New SSH key

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication keys

CPE212 HOME

SHA256: zVcFY96Ey22A9QIWEM1mAHnGaaawsrxYwVbXMHWCWuEc

Added on Aug 25, 2026

Last used within the last week — Read/write

Delete

CPE212

SHA256: oQv3jd0u8V1yCJpH4LZqnEBpstnP1SHUeFrZa/7PAc

Added on Aug 27, 2026

Last used within the last week — Read/write

Delete

CPE212 CentOS WS HOME

SHA256: oqn+WNtQMh1rzkuMyx2XdjKAnFH2nKkgH1gyTCRadqI

Added on Aug 29, 2026

Last used within the last week — Read/write

Delete

centos_chris@localhost:~/CPE212_fegalan — nano homeinventory.yaml

GNU nano 5.6.1

homeinventory.yaml

Modified

```

all:
  children:
    ubuntu:
      vars:
        ansible_user: user123
      hosts:
        server1:
          ansible_host: 192.168.0.101
          os_type: Ubuntu
          apache_package: apache2
          php_package: libapache2-mod-php
        server2:
          ansible_host: 192.168.0.104
          os_type: Ubuntu
          apache_package: apache2
          php_package: libapache2-mod-php

```

Read 18 lines

Help

Write Out

Where Is

Cut

Execute

Location

Undo

Exit

Read File

Replace

Paste

Justify

Go To Line

Redo

GNU nano 5.6.1

install_apache.yaml

```
---
- hosts: all
  become: true
  tasks:

  - name: install apache and php
    package:
      name:
        - "{{ apache_package }}"
        - "{{ php_package }}"
      state: latest
      update_cache: yes
```

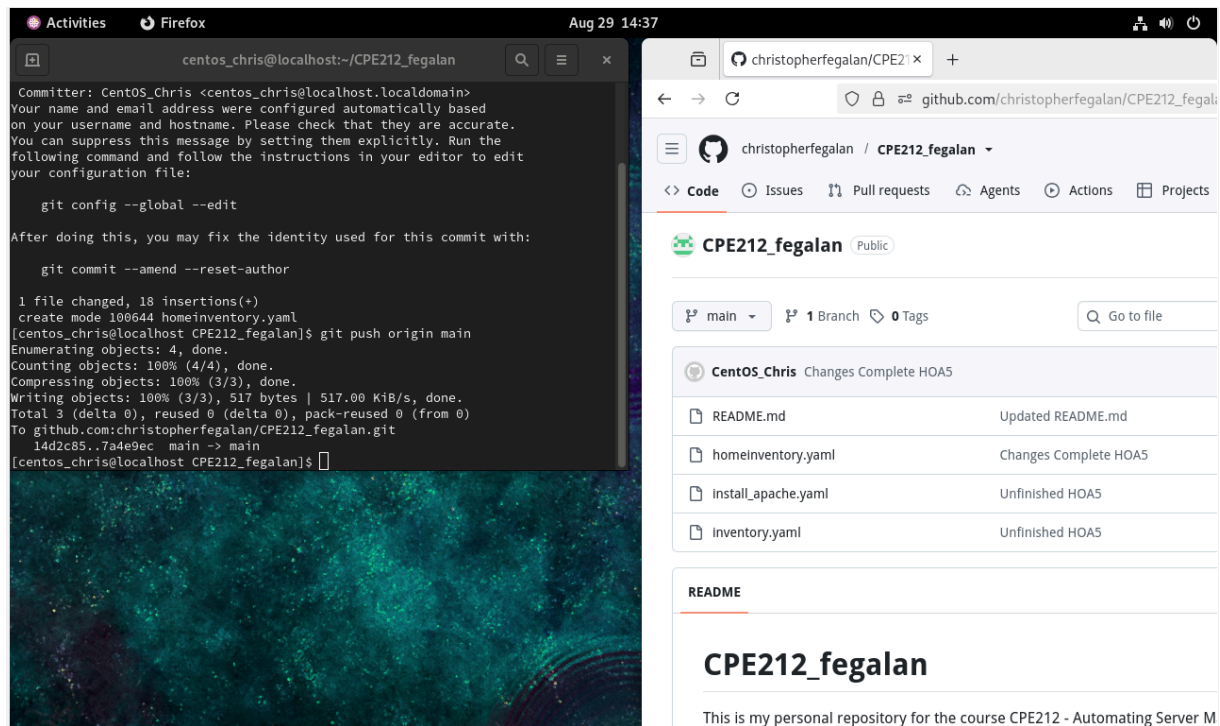
```
[centos_chris@localhost CPE212_fegalan]$ ansible-playbook -i homeinventory.yaml install_apache.yaml -
-ask-become-pass
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]

TASK [install apache and php] *****
changed: [server1]
changed: [server2]

PLAY RECAP *****
server1      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
server2      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
```



- Save for Github

Reflections:

Answer the following:

1. Why do you think refactoring of playbook codes is important?
 - It simplifies reading the code, removes redundancy, and allows us to execute tasks faster and more efficiently.
2. When do we use the “when” command in playbook?
 - For setting conditions in playbooks.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the

permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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